

Sushant Aryal

Geoinformatics and Open Data Engineer

📍 Brugge, Belgium

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• ABOUT ME

Junior Geoinformatics engineer with hands-on experience building Python-based geoprocessing and ETL pipelines, consuming and building web services/REST APIs, and working across both relational and NoSQL data stores. Comfortable working independently against defined procedures, version-controlling code with Git, and collaborating with colleagues. I know open data, FAIR principles, and semantic interoperability from my MSc, and I'm looking to move from theory to practice by applying them in a real open science setting.

• SKILLS

Data Storage: PostgreSQL/PostGIS (RDBMS), MongoDB, spatial database maintenance and topology/geometry checks

Programming: Python, JavaScript/TypeScript, ArcPy, ETL pipelines, batch processing, workflow automation

Geospatial Libraries & Data Pipelines: GDAL, GeoPandas, Shapely, Rasterio, spatial data transformations, CRS handling

Web, Web Services & Standards: Building and consuming REST APIs; OGC standards (WMS/WFS/WPS); GeoServer; GeoJSON, vector tiles; practical understanding of core web architecture

Open Data Standards: Theoretical grounding in FAIR principles and semantic interoperability (MSc coursework), familiarity with the domain's target technologies (RDF, linked-open-data)

Version Control & Containers: Git/GitHub, Docker

Data Quality & Validation: Geometry and attribute validation, nonconformity detection, standardisation against reference datasets

Collaboration: Code review and design discussions with colleagues; client-facing reporting and communication with technical and non-technical stakeholders

• EXPERIENCE

Student Job

June 2026 – July 2026

Earth and Life Institute, Université catholique de Louvain – Louvain-la-Neuve, Belgium

- Applied field boundary delineation model developed in my thesis to generate 2026 agricultural field boundary maps for the Wallonia region
- Processed and fused Sentinel-1 and Sentinel-2 time series data for the Walloon agricultural landscape to support regional-scale delineation

Geospatial AI Internship

May 2025 – Aug 2025

Estonian Land and Spatial Development Board – Tartu, Estonia

- Built reproducible raster processing pipelines (Python, GDAL, Rasterio) ensuring consistent data quality and reliable outputs across multi-dataset workflows.
- Developed automated spatial data validation and preprocessing workflows, reducing manual effort and maintaining data integrity across large orthophoto datasets.
- Worked in a Linux-based, containerised environment applying structured pipeline design for repeatable, audit-ready geospatial processing.
- Conducted large-scale GIS analysis on national aerial imagery datasets, producing spatial outputs to support renewable-energy siting and land-use decision-making.

GIS Developer

May 2023 – Aug 2024

PlantSat – Kathmandu, Nepal

- Configured and maintained GeoServer to publish OGC-compliant spatial services (WMS/WFS), integrating datasets into scalable geospatial workflows across teams
- Designed and managed PostgreSQL/PostGIS databases for large geospatial datasets, implementing spatial indexing, geometry validation, and data quality controls
- Built Python and REST API automation workflows for spatial data validation, standardisation, and delivery of analysis-ready GIS layers to cross-functional teams

- Provided technical guidance to team members on GIS tools, data workflows, and best practices, supporting knowledge sharing across the organisation
- Designed and maintained interactive WebGIS dashboards and web maps (React, OpenLayers, MapLibre GL) to communicate spatial patterns to non-technical stakeholders and support project decision-making
- Delivered GIS analyses on satellite and field data (vegetation indices, spatial aggregation, land classification) to support client reporting and operational planning in agricultural projects

WebGIS Developer Internship

Naxa PVT. LTD – Kathmandu, Nepal

March 2023 – April 2023

- Integrated multi-source spatial datasets (raster, vector, elevation) into a unified mapping platform, ensuring data consistency and quality across inputs
- Developed an interactive web map visualising migrant mobility spatial data, translating client requirements into a functional mapping tool for public-facing use
- Collaborated in a multidisciplinary team to deliver mapping features aligned with project scope and stakeholder needs

• EDUCATION

02/09/2024 – 6/2026

GEO-INFORMATION SCIENCE AND EARTH OBSERVATION FOR ENVIRONMENTAL MODELLING AND MANAGEMENT (M.SC.)

Erasmus Mundus Joint Master (Double degree) (Grade: **Magna Cum Laude**)

University of Tartu, Tartu, Estonia (year 1 2024 - 2025)

Université catholique de Louvain, Louvain-la-Neuve, Belgium (year 2 2025 -2026) — Magna Cum Laude, average: 16.19/20

08/2018 – 05/06/2023 Dhulikhel, Nepal

GEOMATICS ENGINEERING (B.SC) Kathmandu University — Distinction, CGPA 3.71/4.00

• SCHOLARSHIPS/AWARDS

Erasmus Mundus Joint Master Degree Scholarship (2024–2026)

Competitive EU scholarship for the GEM MSc programme in Geo-Information Science and Earth Observation, Track 2: Land Use Systems

Nepal Government Scholarship (2018–2023)

Merit-based 35% tuition scholarship for undergraduate studies, Kathmandu University

• ONLINE PROFILES

LinkedIn: [linkedin.com/in/sushant-aryal-56360b1a4](https://www.linkedin.com/in/sushant-aryal-56360b1a4)

GitHub: github.com/SushantAryal3

• LANGUAGES

English (Proficient C1), French (A1), Dutch (A1- actively improving), Nepali (fluent)